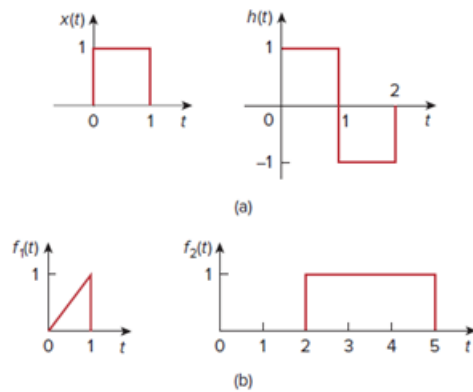


Problem

Obtain the convolution of the pairs of signals in Fig. 15.38.

Figure 15.38:



Step-by-step solution

Step 1 of 8

(a)

From figure 1, $x(t-\lambda)$ and $h(\lambda)$ overlaps in the interval $0 < t < 1$ as follows:

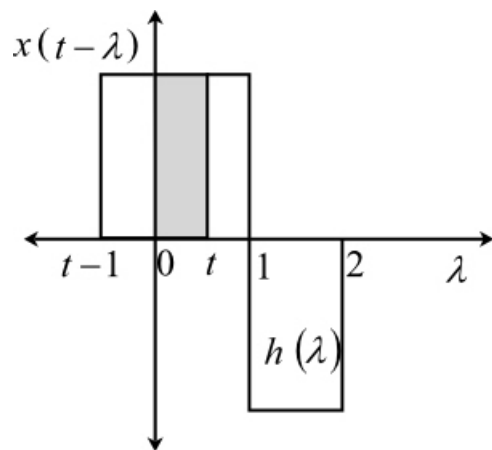


Figure: 1

Step 2 of 8

Calculate $y(t)$ in the interval $0 < t < 1$.

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= \int_0^t (1)(1) d\lambda \\ &= [\lambda]_0^t \\ &= t \end{aligned}$$

From figure 2, $x(t-\lambda)$ and $h(\lambda)$ overlaps in the interval $1 < t < 2$ as follows:

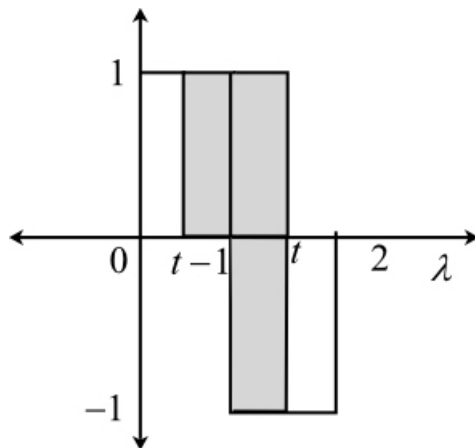


Figure: 2

Step 3 of 8

Calculate $y(t)$ in the interval $1 < t < 2$.

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= \int_{t-1}^1 (1)(1) d\lambda + \int_1^t (-1)(1) d\lambda \\ &= [\lambda]_{t-1}^1 + [-\lambda]_1^t \\ &= 3 - 2t \end{aligned}$$

From figure 3, $x(t-\lambda)$ and $h(\lambda)$ overlaps in the interval $2 < t < 3$ as follows:

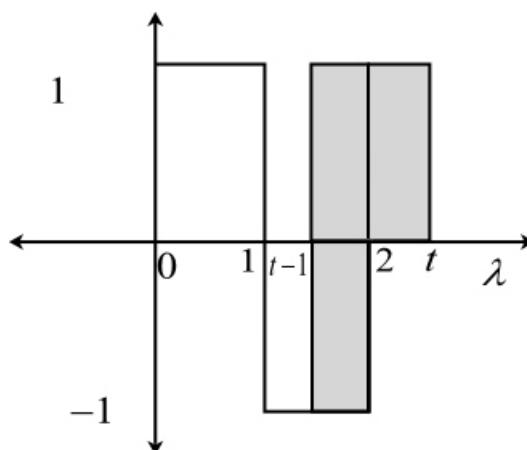


Figure: 3

Step 4 of 8

Calculate $y(t)$ in the interval $2 < t < 3$.

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= \int_{t-1}^2 (1)(-1) d\lambda \\ &= [-\lambda]_{t-1}^2 \\ &= t - 3 \end{aligned}$$

Hence,

$$y(t) = \begin{cases} t & 0 < t < 1 \\ 3 - 2t & 1 < t < 2 \\ t - 3 & 2 < t < 3 \\ 0 & \text{otherwise} \end{cases}.$$

Hence, the required convolution has been determined.

Step 5 of 8

(b)

Determine $f_1(t)$.

$$f_1(t) - 0 = \frac{1-0}{1-0}(t-0)$$

$$f_1(t) = t$$

Now, determine $f_1(t-\lambda)$.

$$f_1(t-\lambda) = t - \lambda$$

For $t < 2$, there is no overlap between $f_1(t-\lambda)$ and $f_2(\lambda)$.

So, $y(t) = 0$.

From figure 4, $f_1(t-\lambda)$ and $f_2(\lambda)$ overlaps in the interval $2 < t < 3$ as follows:

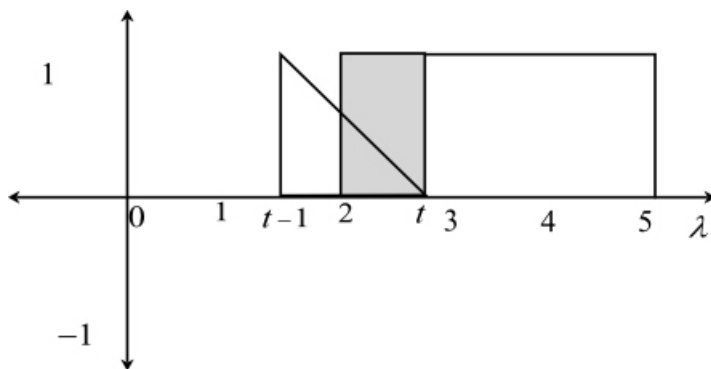


Figure: 4

Step 6 of 8

Calculate $y(t)$ in the interval $2 < t < 3$.

$$\begin{aligned}
 y(t) &= f_1(t) * f_2(t) \\
 &= \int_2^t f_1(t-\lambda) \cdot f_2(\lambda) d\lambda \\
 &= \int_2^t (t-\lambda) \cdot (1) d\lambda \\
 &= \left[\lambda t - \frac{\lambda^2}{2} \right]_2^t \\
 &= \left[t^2 - \frac{t^2}{2} - (2t - 2) \right] \\
 &= \frac{t^2}{2} - 2t + 2
 \end{aligned}$$

From figure 5, $f_1(t-\lambda)$ and $f_2(\lambda)$ overlaps in the interval $3 < t < 5$ as follows:

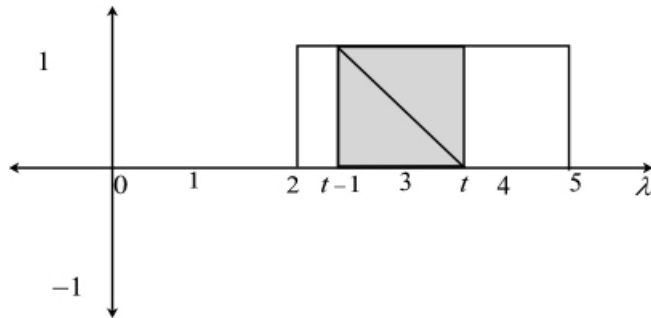


Figure: 5

Step 7 of 8

Calculate $y(t)$ in the interval $3 < t < 5$.

$$\begin{aligned}
 y(t) &= \int_{t-1}^t f_1(t-\lambda) \cdot f_2(\lambda) d\lambda \\
 &= \int_{t-1}^t (t-\lambda) \cdot (1) d\lambda \\
 &= \left[\lambda t - \frac{\lambda^2}{2} \right]_{t-1}^t \\
 &= \frac{1}{2}
 \end{aligned}$$

From figure 6, $f_1(t-\lambda)$ and $f_2(\lambda)$ overlaps in the interval $5 < t < 6$ as follows:

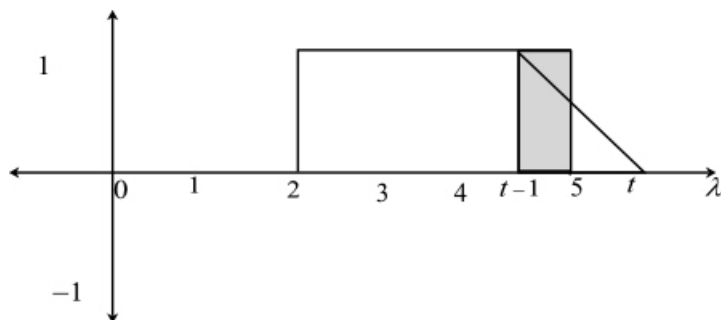


Figure: 6

Step 8 of 8

Calculate $y(t)$ in the interval $5 < t < 6$.

$$\begin{aligned}
 y(t) &= \int_{t-1}^5 f_1(t-\lambda) \cdot f_2(\lambda) d\lambda \\
 &= \int_{t-1}^5 (t-\lambda) \cdot (1) d\lambda \\
 &= \left[\lambda t - \frac{\lambda^2}{2} \right]_{t-1}^5 \\
 &= \frac{-t^2}{2} + 5t - 12
 \end{aligned}$$

Hence,

$$y(t) = \begin{cases} 0 & t < 2 \\ \frac{t^2}{2} - 2t + 2 & 2 < t < 3 \\ \frac{1}{2} & 3 < t < 5 \\ \frac{-t^2}{2} + 5t - 12 & 5 < t < 6 \\ 0 & \text{otherwise} \end{cases}$$

Hence, the required convolution has been determined.